BAGEL-RL: Low-Rank Reinforcement Learning for Interleaved Visual Reasoning

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Abstract

Interleaved multimodal reasoning has recently emerged as a promising paradigm for complex visual reasoning, but existing approaches largely rely on supervised fine-tuning over reasoning trajectories. We investigate whether reinforcement learning can instead efficiently improve interleaved visual reasoning in unified multimodal models. We introduce BAGEL-RL, a parameter-efficient adaptation of BAGEL trained with Group Relative Policy Optimization (GRPO) through Low-Rank Adaptation (LoRA). Our best model improves pretrained BAGEL from 62.49% to 70.36% average performance across visual-reasoning benchmarks, closing 85.5% of the gap to ThinkMorph's 71.70% while updating only 2.6% of the model parameters, and matches or outperforms the fully fine-tuned ThinkMorph on 3 out of 5 benchmarks. These results show that full-parameter fine-tuning is not necessary to obtain strong interleaved multimodal reasoning, and that LoRA-based reinforcement learning provides an effective and efficient alternative.

1 Introduction

Recent progress in unified multimodal models Deng et al. [2025], Chern et al. [2024] has enabled models that treat perception and generation within a single architecture, rather than as separate stages. Building on this capability, recent work Gu et al. [2026], Li et al. [2025] has introduced a new paradigm for visual reasoning in which models interleave visual and textual reasoning steps, generating intermediate visual artifacts (e.g., crops, sketches, or edited regions) as part of the reasoning trace itself. This interleaved approach has shown strong results on tasks requiring fine-grained visual grounding, multi-step visual search, and compositional reasoning over images, where purely text-based chain-of-thought reasoning is insufficient.

Existing approaches to training interleaved multimodal reasoners, however, rely almost exclusively on supervised fine-tuning (SFT) over curated reasoning trajectories, which requires collecting or synthesizing large sets of high-quality interleaved traces, substantially more expensive than collecting single-step visual QA annotations. Reinforcement learning (RL) offers a natural alternative: rather than imitating fixed reasoning trajectories, a model can be trained to directly optimize task reward, potentially discovering more effective reasoning strategies without requiring exhaustive trajectory supervision. Recent work has shown that even low-rank adapters combined with reinforcement learning can enable reasoning behaviors with significant performance gains Morris et al. [2026]. Yet it remains unclear whether reinforcement learning restricted to low-rank updates can enable reasoning capabilities obtained through full-parameter fine-tuning on interleaved multimodal tasks.

In this work, we investigate whether GRPO Shao et al. [2024], applied through LoRA Hu et al. [2021], can efficiently improve interleaved visual reasoning in a unified multimodal model. We introduce BAGEL-RL, a parameter-efficient adaptation of BAGEL Deng et al. [2025] trained with GRPO through LoRA. Starting from an SFT-LoRA warm start, we further optimize the model with

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

GRPO using an exact-match reward on the final answer of each sampled reasoning rollout, updating only 2.6% of the model's parameters throughout.

We evaluate BAGEL-RL on five visual reasoning and perception benchmarks spanning visual puzzle-solving, chart understanding, fine-grained visual search, and compositional perception. Our best model improves average accuracy over the pretrained BAGEL baseline from 62.49% to 70.36%, closing 85.5% of the gap to ThinkMorph, a strong full-parameter fine-tuned baseline, while matching or outperforming it on 3 of 5 benchmarks. These results show that strong interleaved multimodal reasoning gains can be obtained without full-parameter fine-tuning, establishing LoRA-based reinforcement learning as an effective and parameter-efficient alternative.

Our contributions are as follows:

- We show that GRPO-based reinforcement learning applied through LoRA can substantially improve interleaved visual reasoning in a unified multimodal model, without full-parameter fine-tuning.
- We present BAGEL-RL, which improves BAGEL's average benchmark accuracy by 7.87 points, closing the majority of the gap to a strong full-parameter fine-tuned baseline while updating only 2.6% of model parameters.

2 Method

We use a two-stage parameter-efficient training pipeline:

$$\text{BAGEL-7B} \;\rightarrow\; \text{SFT-LoRA} \;\rightarrow\; \text{GRPO-LoRA}.$$

Throughout both stages, the pretrained BAGEL parameters remain frozen and only LoRA parameters are updated (Table 1).

2.1 SFT-LoRA Warm Start

We first fine-tune BAGEL-7B on a subset of interleaved multimodal reasoning trajectories using LoRA, providing the model with the output structure and basic interleaved reasoning behavior required for reinforcement learning. The training objective combines a mean-squared-error loss over generated image latents and a cross-entropy loss over text tokens, weighted 1.0 and 0.5 respectively. The resulting SFT-LoRA checkpoint initializes the GRPO stage.

2.2 GRPO-LoRA

In this stage, we initialize a new set of LoRA adapters (with the same rank) on top of merged checkpoint from the previous stage, which are then optimized via GRPO. For each multimodal prompt x, the policy samples a group of reasoning trajectories $\{y_1, \ldots, y_G\}$, each receiving an exact-match reward based on its final answer:

$$r(y_i) = \begin{cases} 1, & \text{if the final answer is correct and the image step is present,} \\ 0.9, & \text{if the final answer is correct but the image step is skipped,} \\ 0, & \text{if the final answer is incorrect.} \end{cases} \tag{1}$$

We apply a 0.1 image-skip penalty only to otherwise correct rollouts that omit the expected intermediate image generation step. This discourages the policy from abandoning interleaved visual reasoning while leaving incorrect rollouts at zero reward. In practice, we found this constraint important for preventing training collapse into degenerate trajectories that skip visual thoughts.

GRPO computes relative advantages within the sampled group and optimizes a clipped policy objective without a learned value model. Unlike SFT, the reward does not constrain the intermediate reasoning trajectory, allowing different reasoning paths to receive equal reward when they reach the correct answer. We additionally apply a small KL penalty (=0.01) to limit deviation from the reference policy during optimization.

Table 1: Training configuration for each stage. Both stages train LoRA adapters on attention (Q/K/V/O, understanding + generation) and MLP-expert projections across all 28 layers; SFT-LoRA additionally trains the connector and VAE-LLM projection modules.

Stage	Samples	Trainable params	Steps	LoRA: Rank / α	Dropout
SFT-LoRA	10,500	386M (2.59%)	2,500	64 / 128	0.05
GRPO-LoRA	14,490	323M (2.13%)	1,400	64 / 128	0.0

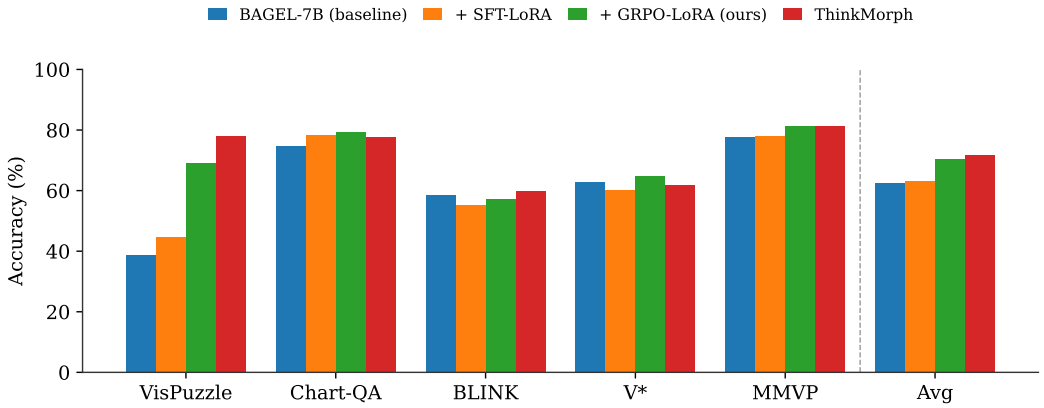


Figure 1: Accuracy (%) across five visual reasoning benchmarks and their average, tracing our training pipeline (BAGEL-7B baseline → SFT-LoRA warm start → GRPO-LoRA) against the fully fine-tuned ThinkMorph.

3 Results and Discussion

3.1 Training Details

We train on ThinkMorph's training data Gu et al. [2026], partitioned into disjoint splits for SFT-LoRA warm-up and subsequent GRPO-LoRA optimization, ensuring reward signals are computed on examples unseen during warm-up. All experiments are run on 8×B100 GPUs (180GB VRAM). SFT-LoRA uses packed sequences (max length 32,768); GRPO-LoRA uses 8 rollouts per prompt with a global batch size of 8. Full configuration details are given in Table 1.

3.2 Evaluation

We evaluate our model on VisPuzzle Gu et al. [2026] (Jigsaw Assembly), ChartQA Masry et al. [2022], BLINK Fu et al. [2024], V* Wu and Xie [2024] (Visual Search), and MMVP Tong et al. [2024], following ThinkMorph's official evaluation script with *gpt-5.4-mini* as judge. We compare against pretrained BAGEL Deng et al. [2025], ThinkMorph Gu et al. [2026], and other open-source vision-language models.

3.3 Discussion

Figure 1 highlights the main trend in our training pipeline. SFT-LoRA provides only a modest improvement over pretrained BAGEL (62.49% → 63.26% average accuracy), while most of the gain appears after reinforcement learning: GRPO-LoRA raises the average to 70.36%, corresponding to a 7.10-point improvement over the SFT warm start and a 7.87-point improvement over the pretrained model. The largest gain is on VisPuzzle, where performance increases from 44.75% after SFT-LoRA to 69.00% after GRPO-LoRA. The relatively small improvement from SFT alone is consistent with its use as a warm start for interleaved reasoning, while the subsequent GRPO stage directly optimizes task success rather than imitation of a fixed reasoning trajectory.

Despite updating only a small fraction of BAGEL's parameters, our final model recovers most of the performance obtained by full-parameter fine-tuning. As shown in Table 2, BAGEL-RL closes 85.5%

Table 2: **Main results.** Accuracy (%) across five visual-reasoning benchmarks. We first fine-tune BAGEL-7B with SFT using LoRA, then initialize GRPO-LoRA training from this SFT checkpoint. ChartQA is averaged over the horizontal- and vertical-bar subsets.

Model	VisPuzzle	ChartQA	BLINK	V*	MMVP	Avg.
BAGEL-7B Deng et al. [2025]	38.75	74.56	58.65	62.83	77.67	62.49
+ SFT-LoRA	44.75	78.17	55.18	60.21	78.00	63.26
+ GRPO-LoRA **(Ours)**	69.00	79.38	57.18	64.92	**81.33**	**70.36**
ThinkMorph Gu et al. [2026]	**78.00**	77.52	59.86	61.78	**81.33**	**71.70**
Qwen3-VL-8B-Thinking Bai et al. [2025]	46.50	**85.70**	59.97	**78.01**	75.67	69.17
InternVL-3.5-8B Wang et al. [2025]	34.25	76.37	**60.76**	69.63	76.00	63.40

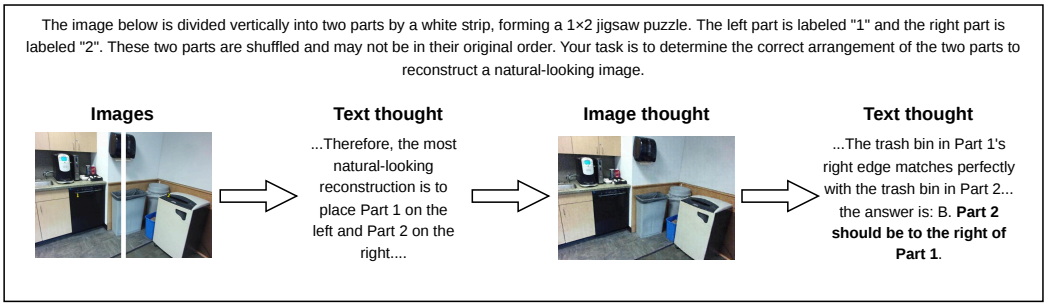


Figure 2: Example of interleaved visual reasoning on a jigsaw-assembly query. Given an image split into two shuffled parts (labeled 1 and 2), the model alternates between textual reasoning and generated visual content: an initial *text thought* hypothesizes the correct part ordering, an *image thought* step generates the corresponding reconstructed image, and a final *text thought* step verifies edge continuity between the two parts (e.g., the trash bin aligning across the join) before committing to an answer.

of the gap between pretrained BAGEL and ThinkMorph in macro-average accuracy, and matches or outperforms ThinkMorph on three of five benchmarks (ChartQA, V*, and MMVP). This suggests that a large fraction of the reasoning improvement obtained by full fine-tuning can be recovered through reinforcement learning restricted to low-rank updates.

We additionally test whether a denser, partial-credit reward improves over exact-match matching. Appendix 5.1 compares our exact-match reward against a partial-credit formulation at a matched step count, and finds the two perform comparably overall (70.36% vs. 70.22% average accuracy), with partial credit offering a modest edge on tasks involving fine-grained visual search or numeric answers. This indicates that GRPO-LoRA's gains are not narrowly dependent on reward shaping, and that a simple exact-match signal is sufficient to realize most of the benefit.

4 Conclusion

We presented BAGEL-RL, a parameter-efficient adaptation of BAGEL for interleaved multimodal reasoning, trained via GRPO-LoRA rather than full-parameter supervised fine-tuning on curated reasoning trajectories. Starting from a brief SFT-LoRA warm start, our best configuration improves average accuracy across five visual-reasoning benchmarks from 62.49% to 70.36%, closing 85.5% of the gap to the fully fine-tuned ThinkMorph, and matching or exceeding it on three of five benchmarks, while updating only 2.6% of model parameters. We further show that this result is not narrowly dependent on reward design: an exact-match reward performs comparably to a denser partial-credit formulation, with stable GRPO training instead depending more on simple safeguards such as a small KL penalty. These results indicate that reinforcement learning can efficiently elicit interleaved visual reasoning in unified multimodal models without the cost of full-parameter fine-tuning, positioning LoRA-based RL as a promising, parameter-efficient route for adapting large unified models to complex visual reasoning tasks.

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5 Appendix

5.1 Exact-Match vs. Partial-Credit Reward

In addition to our main exact-match formulation, we experiment with a partial-credit reward that provides a denser learning signal for near-correct answers. For numeric outputs, the reward varies with the relative difference between the prediction and target; for textual outputs, it assigns partial credit based on string- and word-level overlap. We train this variant for the same 1,400 GRPO steps as our main configuration and report the resulting performance in Table 3.

Table 3: Accuracy (%) of GRPO-LoRA under exact-match vs. partial-credit reward, both at step 1,400. Bold marks the higher value per column.

Reward	VisPuzzle	Chart-QA	BLINK	V*	MMVP	Avg
Exact-match	**69.00**	**79.38**	**57.18**	64.92	**81.33**	**70.36**
Partial-credit	69.25	78.60	56.55	**67.02**	79.67	70.22

The partial-credit variant performs comparably to exact-match training overall, achieving an average accuracy of 70.22% compared with 70.36% for our main configuration. The two rewards lead to somewhat different per-benchmark outcomes: partial credit improves V* from 64.92% to 67.02% and slightly improves VisPuzzle, while exact match performs better on ChartQA, BLINK, and MMVP. These results show that the gains from GRPO-LoRA are not specific to a single binary reward formulation. Given the near-identical overall averages, we adopt exact-match reward as our main configuration on the basis of simplicity, as it requires no reward-shaping design beyond final-answer correctness.